

Thesis Topic

Advanced Techniques in Aspect-Based Sentiment Analysis

Main Idea: Using machine learning models to accurately identify and analyze sentiments related to specific features of products or services.

Importance: Enhances precision in sentiment analysis, improving product insights and customer feedback analysis.

Applications:

- E-commerce: Detailed product reviews.
- Services: Customer feedback on specific service attributes.

Algorithms: Sentiment classification, Natural Language Processing (NLP), deep learning (e.g., BERT, GPT), SVMs (Support Vector Machines)

Tools: Python, NLTK, spaCy, TensorFlow, PyTorch, Scikit-learn, Matplotlib, Seaborn.

Dataset: Online review datasets (kaggle), real time data from review platforms like Amazon, Google Review.

Model Performance Evaluation: Accuracy, precision, and F1-score to evaluate sentiment classification effectiveness.

Outcome: A robust system capable of accurately predicting and categorizing customers sentiments related to distinct features of a product or service.